FNB eWallet to MTN MoMo

UI/UX design specification - screens, controls and interactions

Handoff for the programmer. Covers the banking app (User 1) and the two USSD phone tabs (User 2 eWallet, User 3 MoMo). Source: Implementation Guide v2.1 and the FNB project proposal. All money, accounts and services are simulated.

How to read this document

Every screen has an ID (B01-B07 banking, D00-D10 phones, F00 tester). Each page shows **how the screen looks** on the left, with numbered orange markers on every control, and a table on the right saying **what happens when that control is used**. A "States" table underneath lists the other versions of the screen (errors, loading, expired). Screen IDs, button labels and message texts are final wording unless marked otherwise.

Design principles

Simulation is always visible	Banking header shows a SIMULATION tag and the active user; phones show number, provider and run ID.
Server is the truth	Screens show only what the backend returns. The browser never changes a balance.
One submit, never two	After Submit or PIN the control disables. Uncertain payments say "Do not send again" and offer Check status.
Keep what the user typed	Errors appear beside the field; valid inputs are kept. PINs and codes are never kept or shown.
Words, not colour alone	Every status has a label and an icon shape. Visible labels, clear focus ring, readable text (min 14 px in banking app).
Phones stay plain	The phone screen copies the real FNB USSD pop-up (balance line, numbered choices, Cancel / Send); the banking app may use a modern layout.

Design tokens



Navy #232733 headers, text	**Green #16a34a** primary buttons, success	**Amber #b45309** pending, uncertain	**Red #b91c1c** errors, rejected	**Terminal #10151a** text #7CFC9A, Courier	**Orange #ea580c** spec markers only

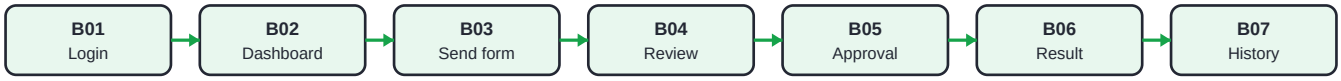
Element	Rule
Typeface	Banking app: system sans-serif (Helvetica/Segoe/Roboto). Phone dialog: the phone's default system font (as in the screenshot), light text on dark grey #202124.
Spacing	8 px grid. Form fields 40 px high, buttons 44 px high (touch friendly), 16 px between fields.
Buttons	Primary = solid green, one per screen. Secondary = outlined navy. Disabled = 50% opacity with spinner or countdown text.
Status chips	SUCCESS = check mark + green; PENDING = clock + amber; UNCERTAIN = question mark + amber; REJECTED/FAILED = cross + red.
Errors	Red text with "(!)" placed directly under the field; focus moves to the first error.
Loading	Buttons show a spinner and are disabled; balances show a grey placeholder bar, never E0 while loading.

Project team

Student no.	Name	Role
202405666	Musawenkosi Magongo	Project Manager
202302550	Setsembele Magagula	Designer
157538	Bandile Lokothwayo	Programmer
202302611	Sambulo Mdluli	Tester

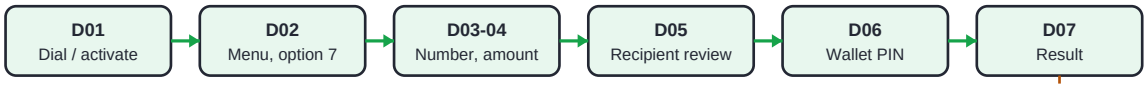
Screen map and navigation

Banking app - User 1



Log out (any screen) -> B01 | B02 History -> B07 | B04 Edit -> B03, Cancel -> B02 | B05 Back -> B04 | B06 Receipt -> B07, Done -> B02

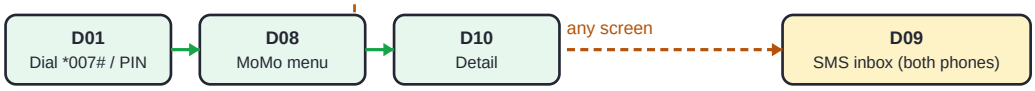
User 2 phone - eWallet (dial *130*392#)



D02 option 5 History -> D10 detail | option 7 Send to MoMo -> D03 | D05 Edit -> D04, Cancel -> D02 | Exit or 120 s idle -> D01

dashed: backend credit notice -> U3 badge + SMS (no phone-to-phone money)

User 3 phone - MoMo



Setup and testing (outside customer screens)



D00 -> opens both phones (D01). F00: scenarios, failures, safe logs, settlement, reset.

Solid green arrows are user actions. The dashed amber line is a backend notice: when MTN saves User 3's credit, the server updates the User 3 tab (unread badge, SMS, balance). Phones never move money to each other directly.

Banking app (User 1)

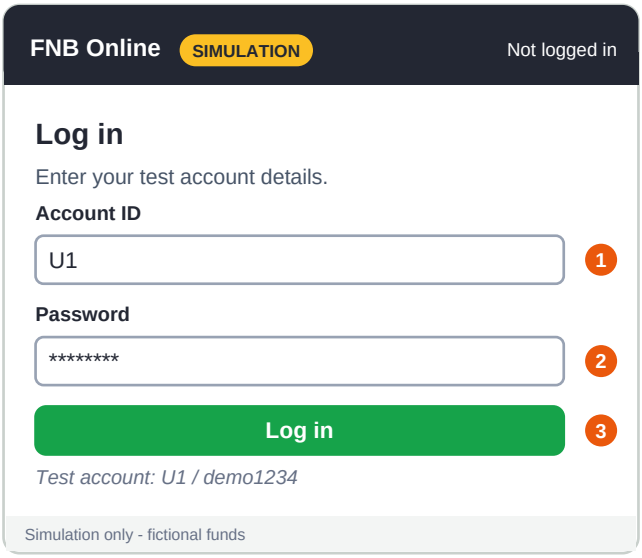
Shared layout on B01-B07

Part	Behaviour
Top bar	"FNB Online" + yellow SIMULATION tag + active user chip (e.g. "User 1 | U1"). Hidden user chip on B01. A Log out link appears on every screen except B01.
Content area	Single column, max 480 px wide, centred on desktop, full width on mobile. Screen title at top; one primary button per screen.
Message tray	Panel beside (desktop) or below (mobile) the content showing User 1 banking messages: approval codes and payment receipts. Newest first, unread dot. Always visible from B05 onward.
Toast / banner	Short result messages above the title. Auto-dismiss after 5 s except errors, which stay until fixed.
Session	15 min inactivity or Log out returns to B01 with "You were logged out". A submitted payment still completes.
Footer	"Simulation only - fictional funds".

B01 Bank login

From: Prototype start | To: B02

User 1 enters a test account ID and password. Never reveals another customer's account.



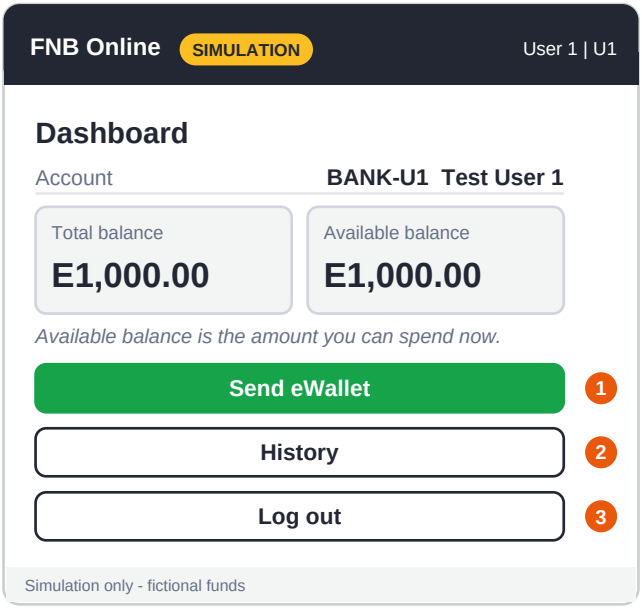
#	Element	When selected / entered
1	Field: Account ID	Text, max 12 characters, trimmed. Empty on Log in: error "Enter your account ID" under the field.
2	Field: Password	Masked. Empty on Log in: error "Enter your password". Never stored in the browser.
3	Button: Log in	Sends ID and password to the backend. Button shows a spinner and is disabled. Correct: opens B02. Wrong: stays on B01, shows "Invalid account ID or password", keeps Account ID, clears password. After 3 failed tries the button locks for 60 s with a countdown.

State	Appearance and behaviour
Default	As shown.
Error	Red message under the password field; Account ID kept.
Locked	Log in button grey: "Try again in 0:42". Counts down then re-enables.
Loading	Spinner in button; fields disabled.

B02 Dashboard

From: B01 login, B06, B07 | To: B03, B07, B01

Shows the account and both balances. Starting point for sending money.



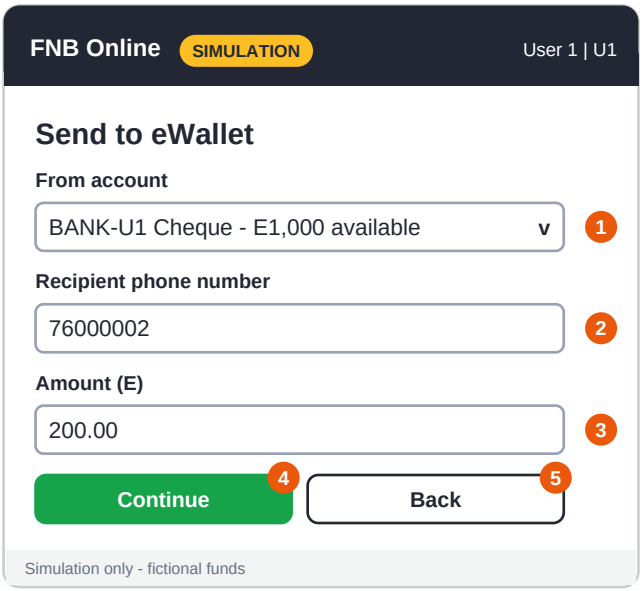
#	Element	When selected / entered
1	Button: Send eWallet	Opens B03 Send form.
2	Button: History	Opens B07 payment list.
3	Button: Log out	Ends the session and opens B01. Any payment already submitted still completes.

State	Appearance and behaviour
Loading	Both balance cards show grey bars; buttons stay enabled.
After a payment	Balances refresh from the backend (E1,000 becomes E800 after S1-001).
Backend unavailable	Banner "Balances unavailable. Try again" with a Retry link; Send eWallet disabled.

B03 Send form

From: B02, B04 Edit | To: B04, B02

User 1 chooses the source account, enters User 2's phone number and the amount.



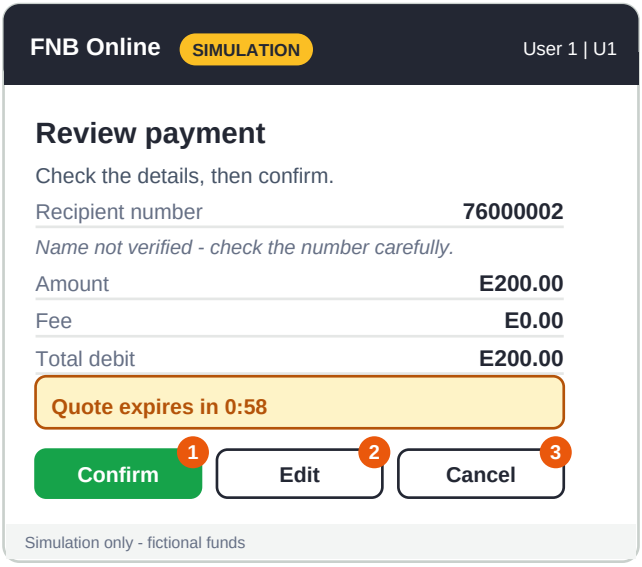
#	Element	When selected / entered
1	Field: From account	Dropdown of accounts owned by the logged-in user only, each with available balance. Default BANK-U1. Browser never sends an owner ID as proof.
2	Field: Recipient phone number	Numeric keypad. 8 digits; spaces ignored. Invalid: "Enter an 8-digit number" under the field. No name lookup; the number is shown back on B04.
3	Field: Amount (E)	Positive, max 2 decimals, not above available balance or the E10,000 limit. Errors: "Enter an amount above E0", "More than your available balance", "Above the E10,000 limit". Errors post nothing.
4	Button: Continue	Validates all fields. If valid, asks the backend for a quote (fee and 60 s expiry) and opens B04. Invalid: stays here, keeps every value, focus moves to the first error.
5	Button: Back	Returns to B02. Typed values are discarded.

State	Appearance and behaviour
Error	Red text under each affected field; entered values stay.
Insufficient funds	Amount error shown; Continue still enabled so the user can correct it.
Backend unavailable	Banner "Cannot get a quote now. Try again."

B04 Payment review

From: B03 | To: B05, B03, B02

Shows the saved quote exactly as it will be approved. The entered number is always displayed; a name appears only if verified.



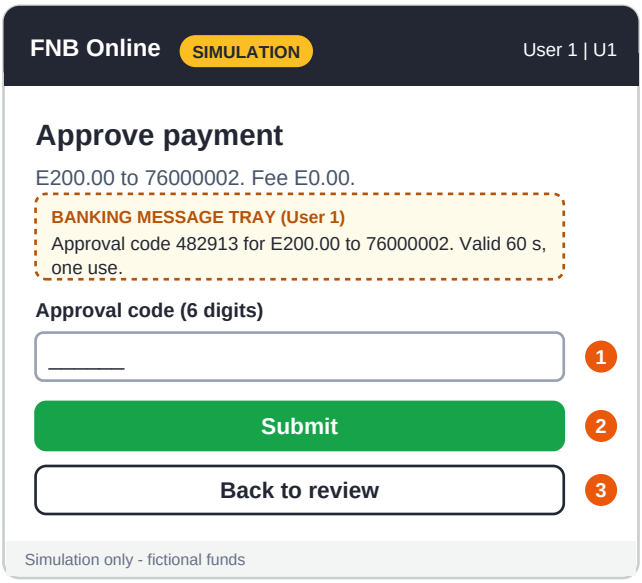
#	Element	When selected / entered
1	Button: Confirm	Generates a one-use approval code, puts it in the message tray, opens B05.
2	Button: Edit	Opens B03 with values filled in. The old quote is discarded.
3	Button: Cancel	Opens B02. Nothing is sent or saved.

State	Appearance and behaviour
Quote expired	Banner turns red "This quote expired". Confirm is replaced by "Review again", which requests a fresh quote.
Amount or fee changed	Return here with the new total; approval must be repeated.
Name verified	If the backend verifies the holder, show name under the number; otherwise the "not verified" note.

B05 Payment approval

From: B04 | To: B06, B04, B03

User 1 enters the one-use test code from the banking message tray.



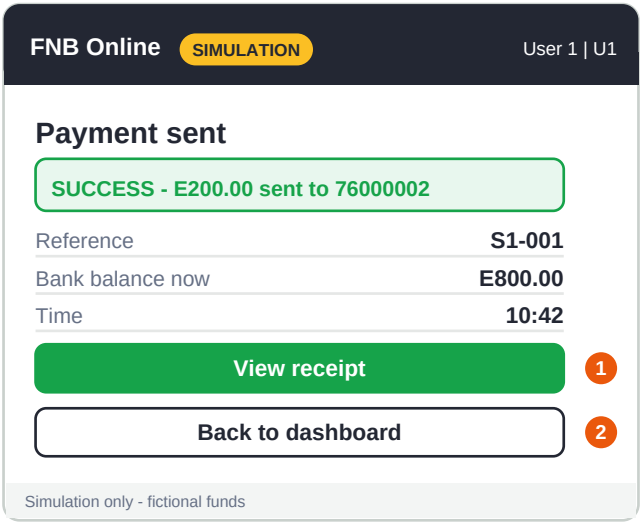
#	Element	When selected / entered
1	Field: Approval code (6 digits)	Numeric keypad, 6 digits, autocomplete off. Wrong code: "Wrong code. 2 tries left" under the field. Code works only for this quote.
2	Button: Submit	Sends the code once; button disables with a spinner (no repeat clicks). Correct: opens B06. Wrong: error, stays. 3 wrong: payment cancelled, opens B03. Expired: opens B04 with "Code expired".
3	Button: Back to review	Opens B04. The code stays valid until its expiry.

State	Appearance and behaviour
Waiting	Spinner on Submit; screen text "Processing..."; Back disabled.
Wrong code	Red message under field; code field cleared; counter of tries left.
Expired	Banner "Code expired" and automatic return to B04.

B06 Result

From: B05 | To: B07, B02

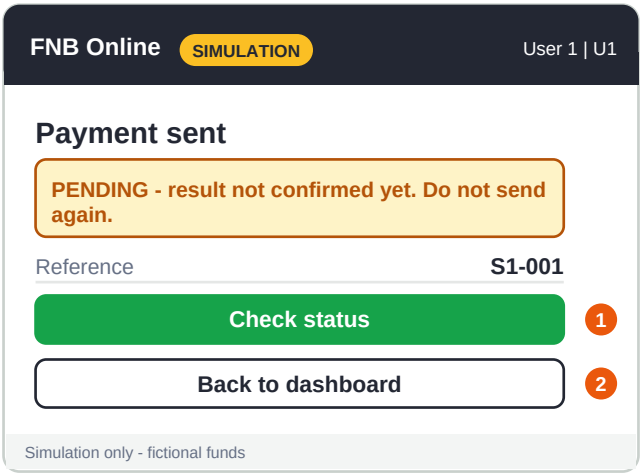
Shows the outcome. SUCCESS only after the bank debit and eWallet credit are saved.



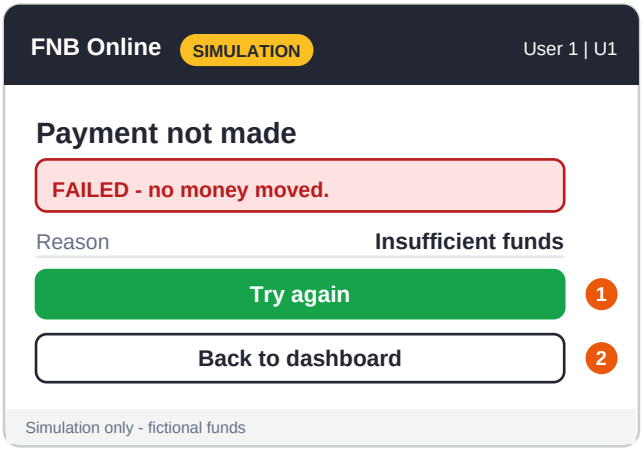
#	Element	When selected / entered
1	Button: View receipt	Opens the receipt detail in B07.
2	Button: Back to dashboard	Opens B02 with refreshed balances.

State	Appearance and behaviour
SUCCESS	Green banner with check icon, reference and new balance (shown).
UNKNOWN	Amber banner "Result not confirmed yet. Do not send again." Button "Check status" reloads the status by payment ID; no resend button exists.
FAILED	Red banner "Payment not made. No money moved." Button "Try again" opens B03 with values filled.

B06 variant: UNKNOWN



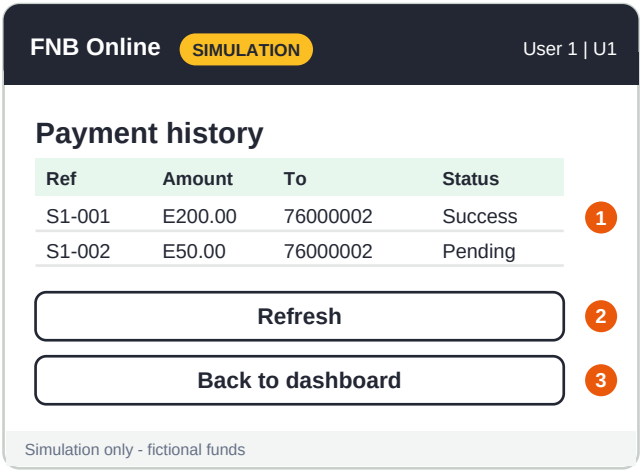
B06 variant: FAILED



B07 History / detail

From: B02, B06 | To: B02

Recent payments first with type, amount, recipient, status and reference.



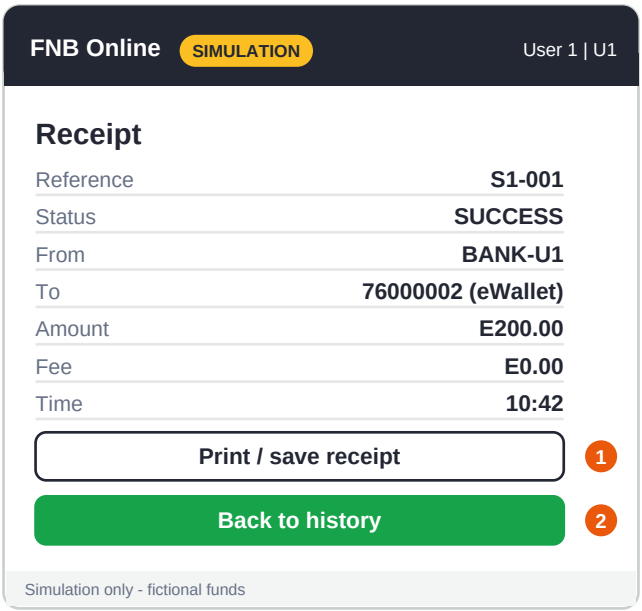
#	Element	When selected / entered
1	Row: any list row	Selecting a row opens the receipt detail (below). Newest first. List loads saved records from the backend.
2	Button: Refresh	Reloads the list from saved records.
3	Button: Back to dashboard	Opens B02.

State	Appearance and behaviour
Empty	Text "No payments yet" and a Send eWallet button.
Receipt detail	Opens in place (see below). Status chip uses text plus icon.

B07 Receipt detail

From: B07 list | To: B07 list

Read-only receipt for one payment.



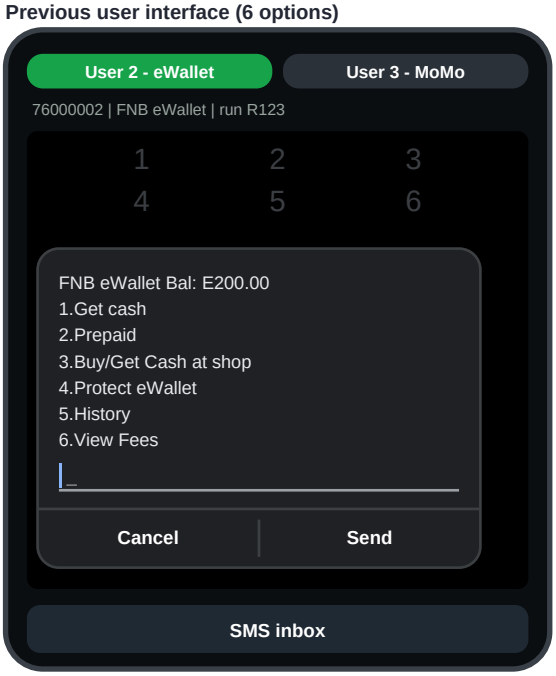
#	Element	When selected / entered
1	Button: Print / save receipt	Opens the browser print dialog for this receipt only.
2	Button: Back to history	Returns to the list.

Phone tabs (User 2 eWallet and User 3 MoMo)

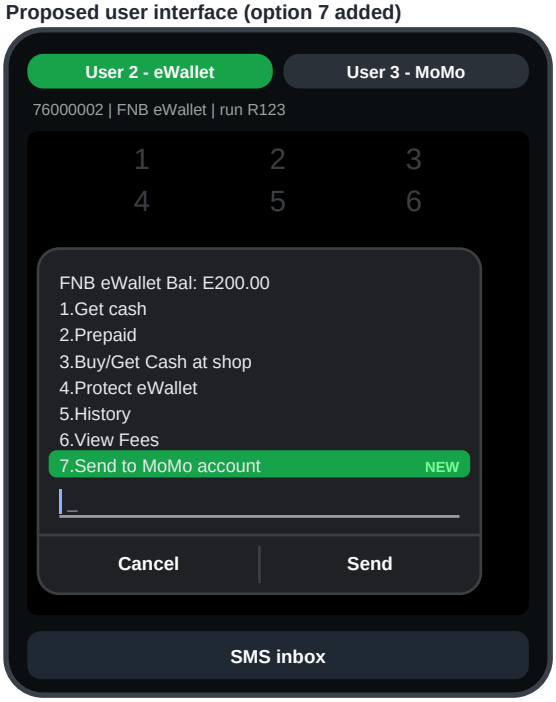
The phone screen copies the real FNB eWallet USSD dialog: a dark pop-up with the balance on the first line, numbered choices, a reply line, and Cancel / Send buttons. The only change is the new option **7.Send to MoMo account**. The workspace adds two tabs at the top and an SMS inbox button under the dialog.

Previous interface and proposed interface

Previous user interface (6 options)



Proposed user interface (option 7 added)

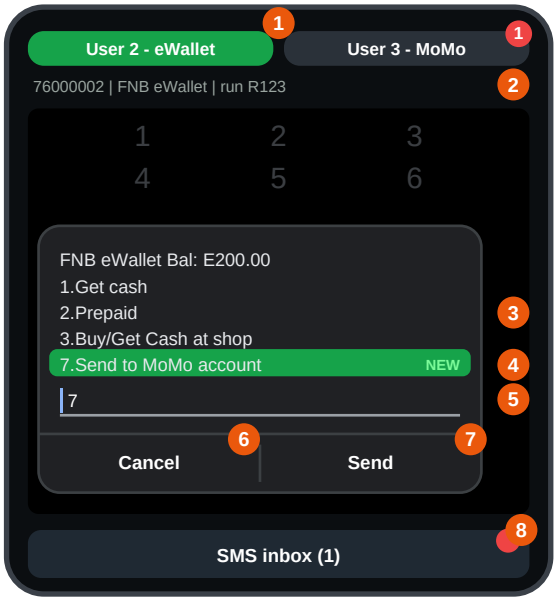


Starting point: User 1 has sent E200 to User 2 in the banking app, so User 2's eWallet pocket shows **Bal: E200.00**. User 2 then dials *130*392#, enters the wallet PIN, and chooses **7** to send part of that money (E150) to User 3's MoMo wallet. After the transfer the first line reads **Bal: E50.00**.

Phone frame Same on both tabs

From: - | To: -

Dialog, reply line, buttons and tabs.



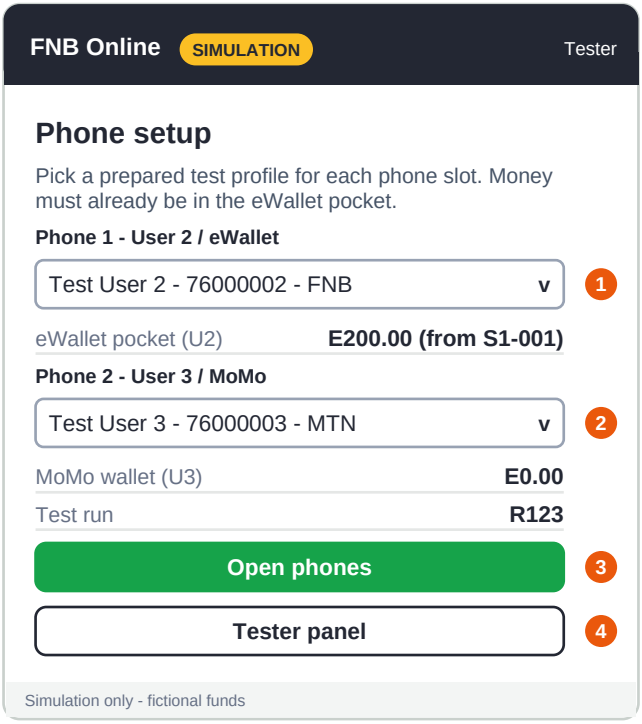
#	Element	When selected / entered
1	Tabs (top)	"User 2 - eWallet" and "User 3 - MoMo". Switching changes only which phone is visible. The hidden phone keeps its screen, input and session; its SMS badge keeps updating.
2	Header line	Number, provider and run ID. Always visible.
3	Options: Options 1-6	Numbered choices are typed in the reply field, then Send.
4	Option: 7.Send to MoMo account	Proposed new choice.
5	Field: 7	Single-line reply field. Masked (*****) on PIN steps. Valid choices are re-shown after an invalid reply.
6	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
7	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.
8	Button: SMS	Opens the SMS inbox (D09) for this phone. Red badge = unread messages. Works even when no dialog is open.

Rules: the dialog closes after 120 s of inactivity (unfinished inputs and login cleared); the reply field empties after every Send; typing 0 goes back one step wherever "0.Back" is shown; PINs are never echoed or saved in history.

D00 Phone setup

From: Prototype start | To: D01, F00

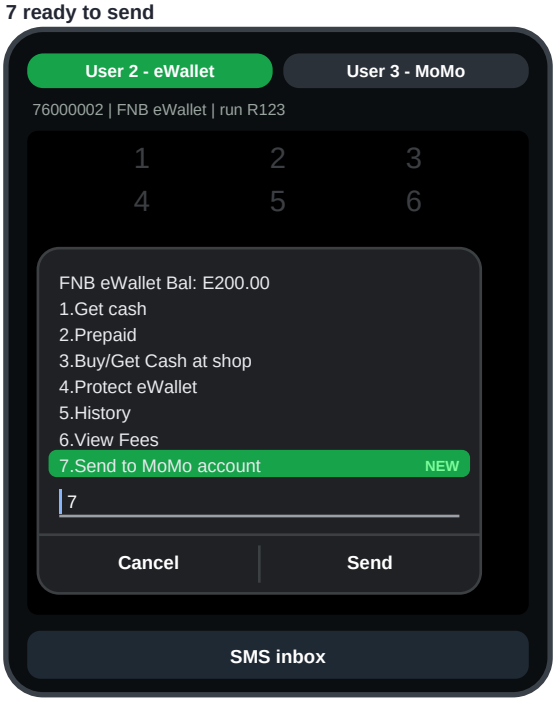
Where each phone slot gets its test profile. The kv rows show the pocket balances loaded from the backend (read-only, never editable here).



#	Element	When selected / entered
1	Field: Phone 1 - User 2 / eWallet	Choose a prepared fictional profile. Shows number and provider. Selecting does not authorise a payment or skip activation. Changing it ends that phone's session.
2	Field: Phone 2 - User 3 / MoMo	Same rules. Slot 1 is always the eWallet phone, slot 2 the MoMo phone.
3	Button: Open phones	Opens the two-tab workspace. Both phones start at D01. User 2 then sees the dialog on the right.
4	Button: Tester panel	Opens F00 in a separate panel.

State	Appearance and behaviour
Pocket empty (E0)	Banner "User 2 has no money yet. Send from the banking app first" ; Open phones still works but option 7 will refuse with "Insufficient available funds".
Pocket funded	As shown: E200.00 appears after User 1 completes B06.

What User 2 sees after Open phones, dial, PIN: option 7 ready to send

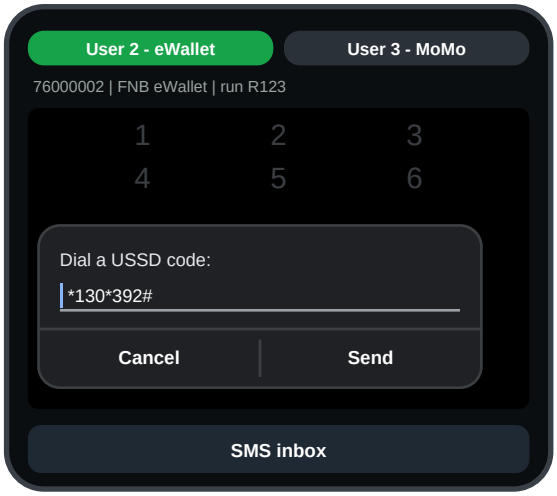


D01 Dial, activate and sign in

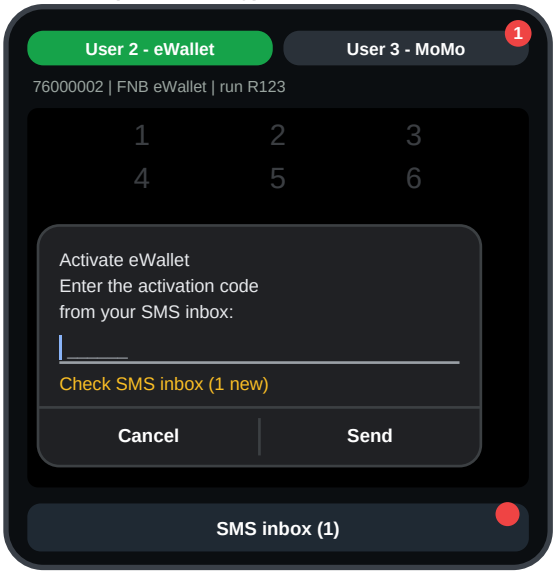
From: D00, D02 Cancel | To: D02 / D08

First screens on each phone. User 2 activates once; afterwards each phone asks only for its own PIN.

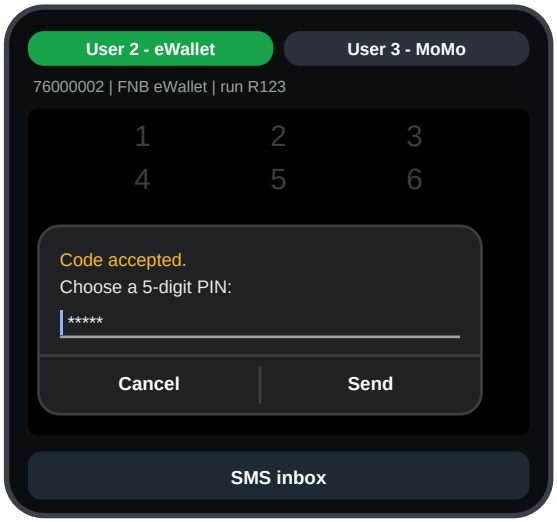
1. Dial



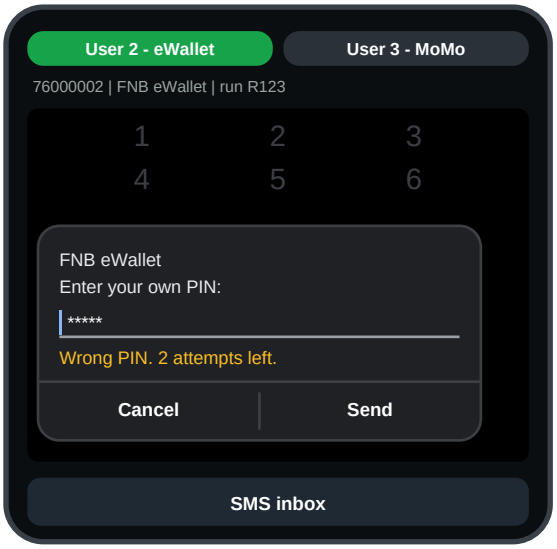
2. Activate (first time only)



3. Choose PIN



4. Enter PIN

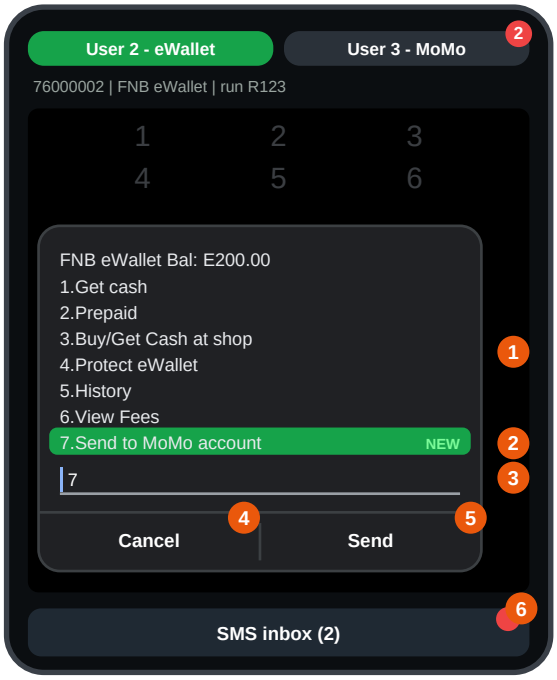


Step	User input	Result
1 Dial	User 2: *130*392#. User 3: *007#.	Right code, eWallet not activated: step 2. Right code and activated (or MoMo): step 4. Wrong code: "Invalid code", stays.
2 Activate	6-digit code from this phone's SMS inbox	Correct: step 3. Wrong or expired: "Wrong or expired code", stays. One use only.
3 Choose PIN	Exactly 5 digits (masked)	Valid: "eWallet activated", then D02. Invalid: "PIN must be 5 digits". Activation does not create a bank account.
4 Enter PIN	Own 5-digit PIN (masked)	Correct: D02 (eWallet) or D08 (MoMo). Wrong: "Wrong PIN. N attempts left". 3 wrong: "Too many attempts. Wait 60 s", input locked.

D02 eWallet menu (User 2)

From: D01, D07 Home | To: D03, D10, D01

The main menu after PIN. First line shows the live pocket balance from the backend.



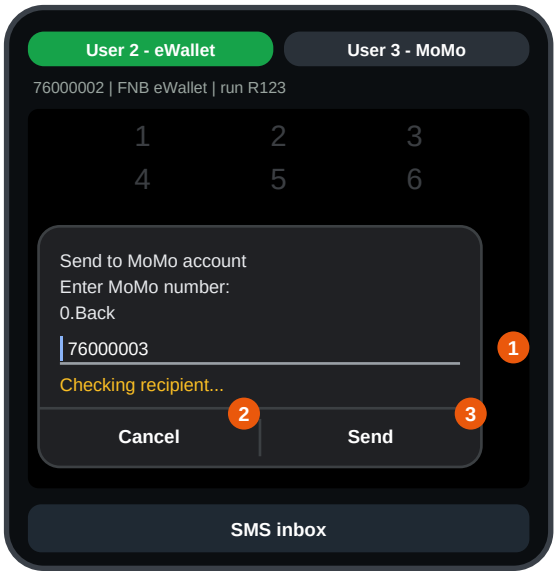
#	Element	When selected / entered
1	Options: Options 1-6	Existing FNB items. In this simulation: 5 opens recent transfers (D10 list); 6 shows "MoMo transfer fee: E0 (test)"; 1, 2, 3 and 4 show "Not part of this simulation" and the menu returns.
2	Option: 7.Send to MoMo account	PROPOSED. Type 7 then Send. Opens D03 to enter the MoMo number. This is how User 2 sends money from the eWallet pocket to MoMo. Needs an activated wallet and the tester setting for the proposed option switched on.
3	Field: 7	Type 1 to 7 then Send. Anything else: "Choose 1 to 7" and the menu repeats.
4	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
5	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.
6	Button: SMS	Opens the SMS inbox (D09) for this phone. Red badge = unread messages. Works even when no dialog is open.

State	Appearance and behaviour
Balance line	E200.00 after S1-001; E200.00 with E150 on hold shows "Bal: E50.00" (available); E50.00 after the final debit.
Option 7 hidden	If the tester setting is off the menu shows 1-6 only.
Idle 120 s	Dialog closes: "Session ended. Dial again."

D03 MoMo recipient number

From: D02 option 7 | To: D04

User 2 enters the receiving MoMo number.

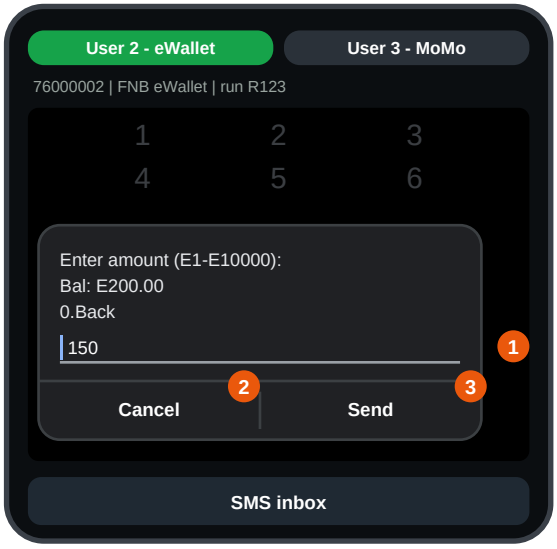


#	Element	When selected / entered
1	Field: 76000003	8 digits. Shows "Checking recipient..." while the simulated MTN lookup runs. Eligible: opens D04. Not found or inactive: "Recipient cannot receive funds. Nothing reserved." stays here. Lookup timeout: "Verification unavailable. Try later." (never says inactive). Typing 0 returns to D02.
2	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
3	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.

D04 Amount

From: D03, D05 Edit | To: D05

User 2 enters how much of the pocket money to send.

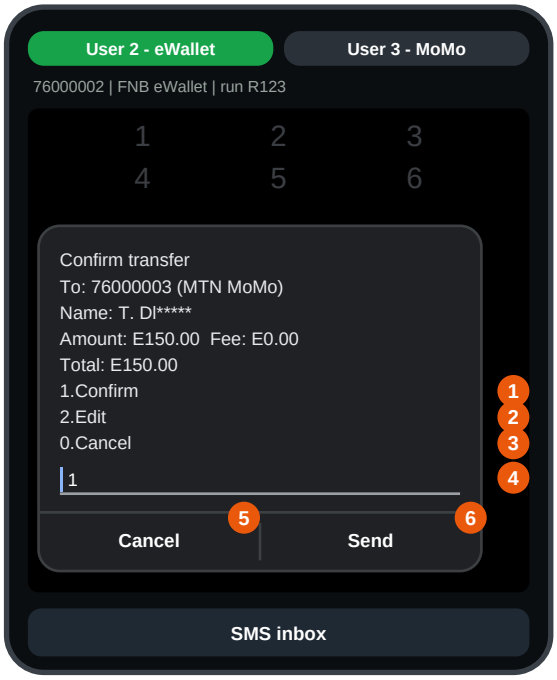


#	Element	When selected / entered
1	Field: 150	Positive, max 2 decimals, within available funds and per-transfer and daily limits. Errors stay on this step: "Insufficient available funds", "Enter a valid amount", "Above transfer limit". Valid: opens D05 (quote saved for 60 s). 0 returns to D03.
2	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
3	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.

D05 Recipient review

From: D04 | To: D06, D04, D02

Shows the full entered number, provider, partly hidden name only if permitted, amount, fee and total.



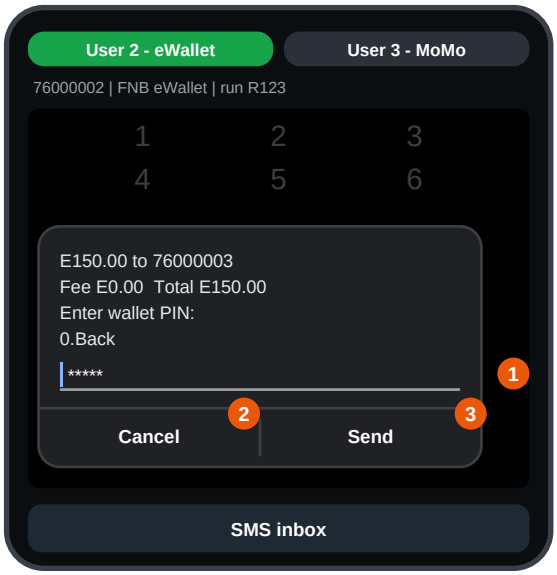
#	Element	When selected / entered
1	Option: 1.Confirm	Opens D06 PIN entry.
2	Option: 2.Edit	Opens D04; the old quote is replaced when a new amount is entered.
3	Option: 0.Cancel	Opens D02. Nothing reserved or sent.
4	Field: 1	Type 1, 2 or 0 then Send.
5	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
6	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.

State	Appearance and behaviour
Name unavailable	Line "To: 76000003" without a name; payment may continue if checks pass.
Quote expired (60 s)	"Quote expired. Review again." returns to D04.
Verification unavailable	"Cannot verify recipient. Try later." Payment blocked.

D06 Wallet PIN

From: D05 | To: D07, D05

Approves exactly the summary shown.



#	Element	When selected / entered
1	Field: *****	Own 5-digit PIN, masked, never saved in history. Correct: submits ONE request and opens D07 (PENDING). Wrong: "Wrong PIN. N attempts left", nothing reserved. 3 wrong: 60 s lock. 0 returns to D05.
2	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
3	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.

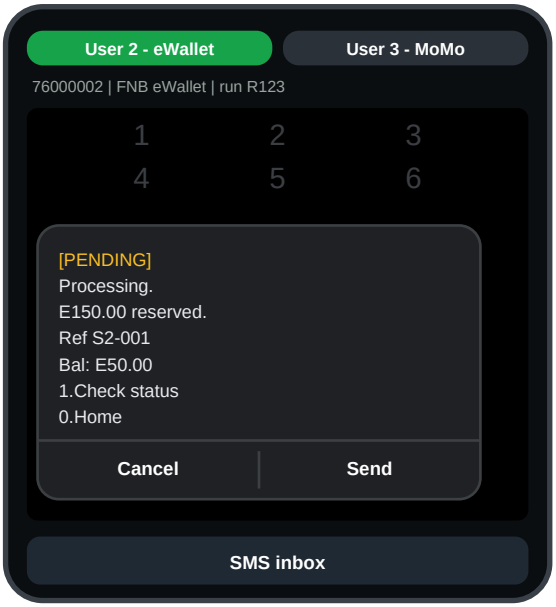
State	Appearance and behaviour
Locked	"Too many attempts. Wait 60 s." Input disabled with countdown.
Submitting	Reply field disabled; "Sending..."; Send disabled to stop repeats.

D07 Payment result (4 versions)

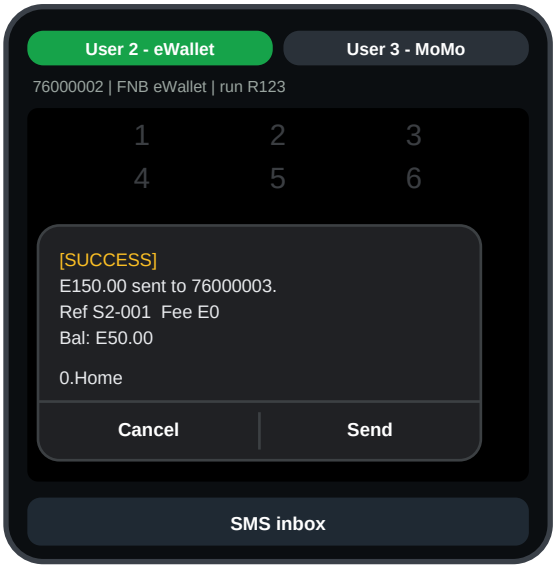
From: D06 | To: D02 via Home

Live status of the payment. Updates in place; no new payment can start from here.

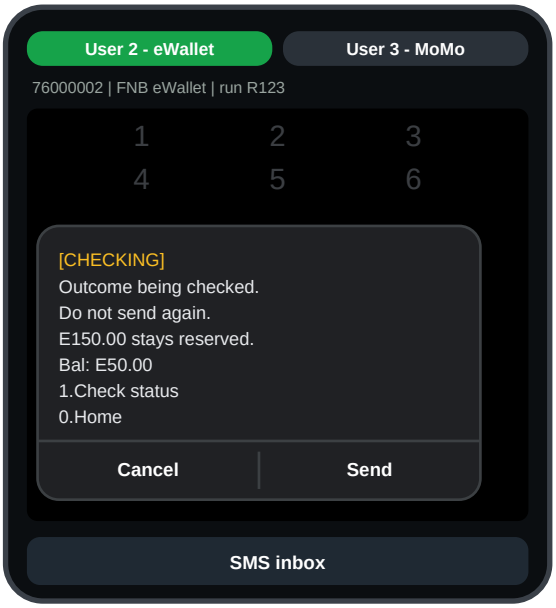
Version: PENDING



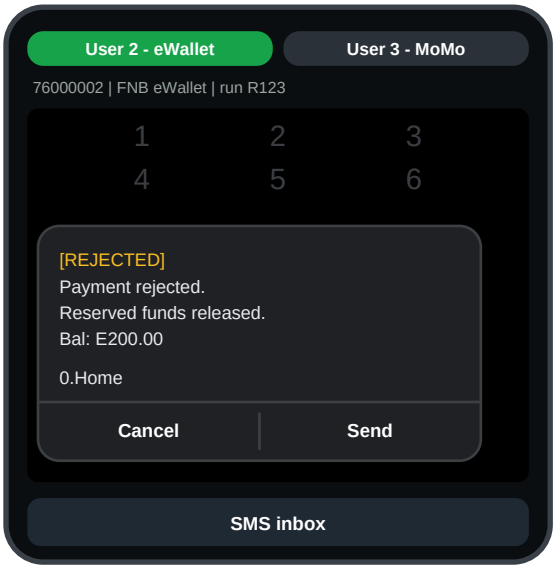
Version: SUCCESS



Version: UNCERTAIN



Version: REJECTED

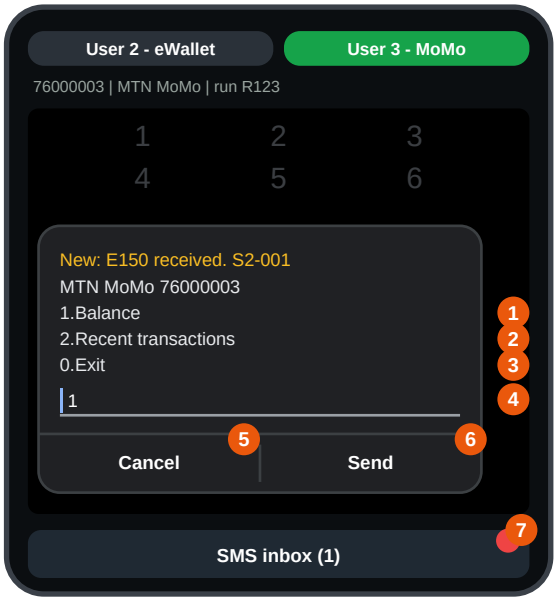


#	Element	When selected / entered
1	Option: 1.Check status	Shown for PENDING and UNCERTAIN. Asks the backend for the status of the original payment ID and refreshes. Never creates a new payment.
2	Option: 0.Home	Opens D02. Closes this screen only; never cancels or reverses a payment. The payment stays in History (option 5).
3	Automatic refresh	About every 2 s while open; PENDING after 5 s adds "Still processing". Unresolved after 5 min: "Contact support" with the reference.
4	Messages	PENDING "Processing. E150 reserved." | UNCERTAIN "Outcome being checked. Do not send again." | REJECTED "Payment rejected. Reserved funds released." (only after release is saved) | SUCCESS "E150 sent to [number]. Ref [ID]."

D08 MoMo menu (User 3)

From: D01 (own PIN) | To: Balance, D10, D01

User 3 sees a credit only after MTN has saved it. A yellow notice line appears when a new credit arrives.

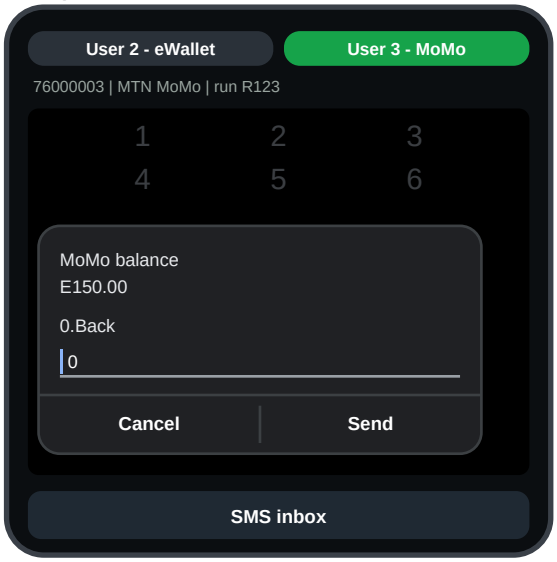


#	Element	When selected / entered
1	Option: 1.Balance	Shows "MoMo balance E150.00" from the backend. 0 returns.
2	Option: 2.Recent transactions	Lists received credits, e.g. "+E150 from 76000002 S2-001"; typing its number opens D10.
3	Option: 0.Exit	Ends the session, returns to D01.
4	Field: 1	Type 1, 2 or 0 then Send. Receiving funds needs no PIN or approval from User 3.
5	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
6	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.
7	Button: SMS	Opens the SMS inbox (D09) for this phone. Red badge = unread messages. Works even when no dialog is open.

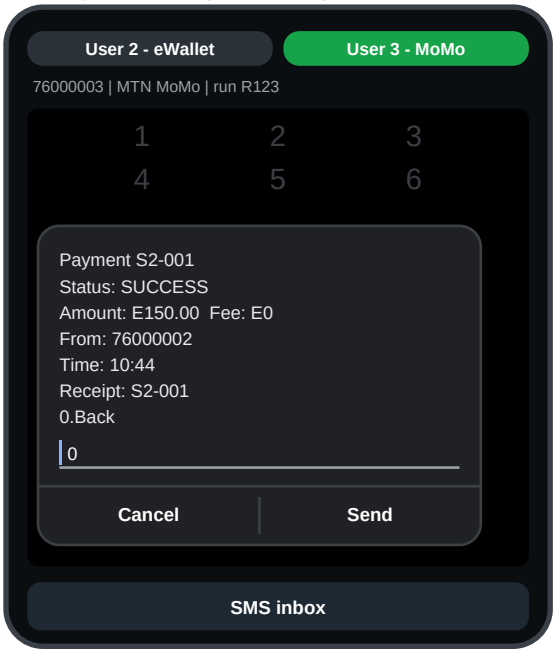
State	Appearance and behaviour
Credit arrives mid-use	Unread badge on SMS inbox updates; the balance refreshes next time Balance is opened. The current screen is never replaced mid-input.
Offline	"Offline - balances may be outdated"; Send disabled.

D10 Payment detail

D08 option 1: Balance



D10: Payment detail (User 3 view)

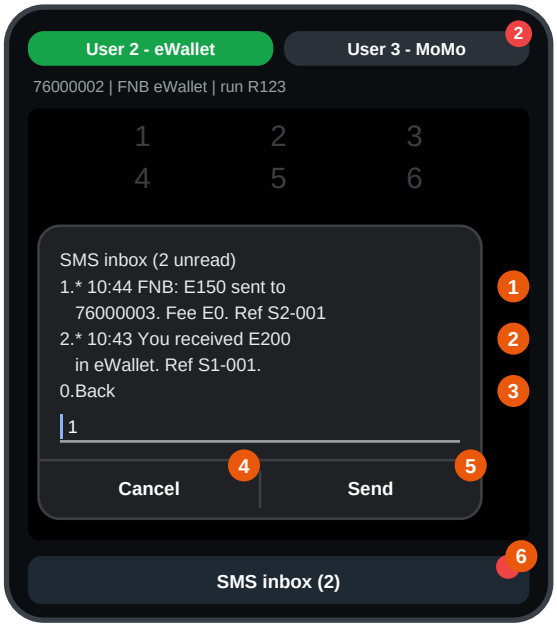


D10 opens from History (option 5, User 2) or Recent transactions (User 3). User 2 sees their own eWallet balance and the debit; User 3 sees their own MoMo balance and the credit. 0 returns to the list. An SMS alone is never proof of payment status; the screen reloads the status from the backend.

D09 SMS inbox

From: Any screen (SMS inbox button) | To: Previous screen

Sender, text, time and payment reference. Works even with no dialog open. Never contains PINs or codes.



#	Element	When selected / entered
1	Option: 1.* 10:44 FNB: E150 sent to	Type a message number then Send. Opens the full text, marks it read and lowers the badge.
2	Option: 2.* 10:43 You received E200	Same behaviour. Unread messages show "*".
3	Option: 0.Back	Returns to the previous menu position.
4	Button: Cancel	Closes the USSD dialog and ends this phone's session: unfinished inputs and login are cleared and the phone returns to D01. A payment already submitted is not cancelled.
5	Button: Send	Submits the typed reply to the current step. Enter key does the same. An empty reply does nothing.
6	Button: SMS	Opens the SMS inbox (D09) for this phone. Red badge = unread messages. Works even when no dialog is open.

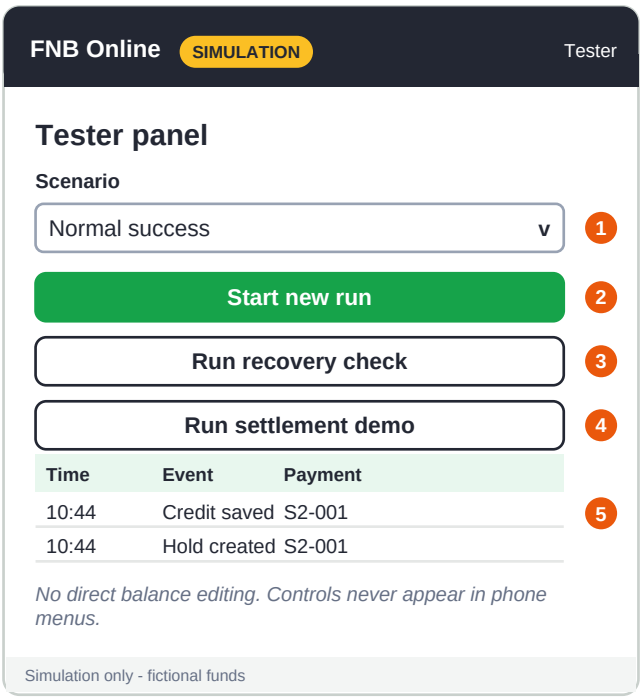
State	Appearance and behaviour
Empty	"No messages".
New message while open	Added at the top with "*"; badge unchanged because the inbox is open.
Per phone	U2 gets: Stage 1 credit, activation code, final debit, rejection. U3 gets: MoMo credit message.

Tester panel and message catalogue

F00 Tester panel

From: D00 / any time | To: Same screen

Separate technical controls, kept out of customer menus.



#	Element	When selected / entered
1	Field: Scenario	Choose normal success, inactive recipient, late refusal, or reply lost after credit (UNCERTAIN). Applies to the next phone transfer.
2	Button: Start new run	Creates a new run ID and clears both phones and inboxes. Old replies cannot affect the new run.
3	Button: Run recovery check	Re-checks uncertain payments by original ID. Completes the debit if a credit was saved; releases funds only after a final no-credit record.
4	Button: Run settlement demo	Settles accepted amounts in a batch (UNSETTLED to SETTLED). Customer balances do not change.
5	Row: any list row	Safe log lines only: no PINs, passwords or codes. Newest first.

Message catalogue (final wording)

Where	Message
USSD success	E150 sent to [number]. Ref [ID].
USSD pending	Processing. E150 reserved.
USSD uncertain	Outcome being checked. Do not send again.
USSD rejected	Payment rejected. Reserved funds released.
SMS U1 (tray)	E200 sent to [U2 number] via eWallet. Ref S1-001.
SMS U2	You received E200 in eWallet. Ref S1-001. | E150 sent to [U3 number]. Fee E0. Available E50. Ref S2-001.
SMS U3	MoMo: received E150. Balance E150. Ref S2-001.
SMS U2 rejected	Payment S2-001 rejected. E150 reservation released.
Offline	Offline - balances may be outdated.
Recipient check timeout	Verification unavailable. Try later.

Notes for the programmer

1. The dial codes *130*392# (FNB eWallet) and *007# (MTN) come from the sources; the menu numbers, PIN rules, "Send to MoMo" option and all limits are chosen for this simulation. 2. Marker numbers in the tables follow the order of the controls on screen. 3. Wording of every error is final; show it beside the field (banking) or as a yellow line above the prompt (phones). 4. Test amounts: E200 then E150; final balances E800 / E50 / E150.